

# Dependence Uncertainty

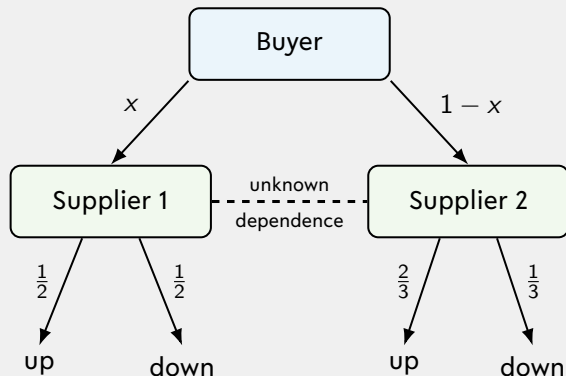
Gerrit Bauch<sup>1</sup>    Lorenz Hartmann<sup>2</sup>

<sup>1</sup>IMW, Bielefeld University

<sup>2</sup>Basel University

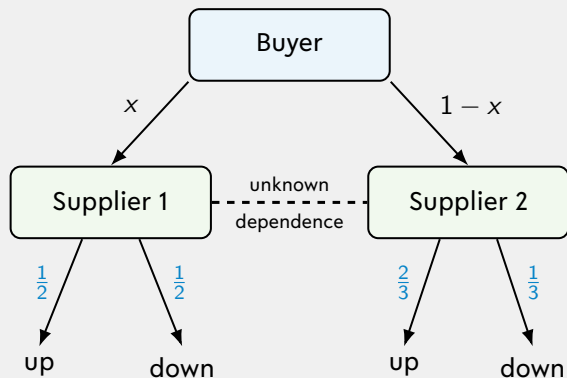
IFORS Vienna, July 2026

# Dual-sourcing procurement with disruption



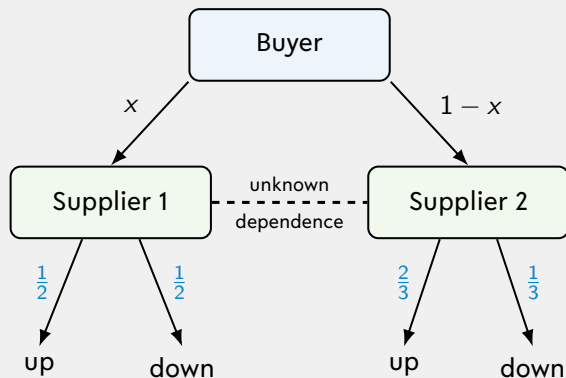
- Sourcing share  $x$  to Supplier 1.
- Known individual disruption rates (audits, service histories).
- Dual sourcing enhances reliability and resilience, in case of common shocks (natural disaster, geopolitics).
- Optimal sourcing depends on the unknown joint disruption structure.

# Dual-sourcing procurement with disruption



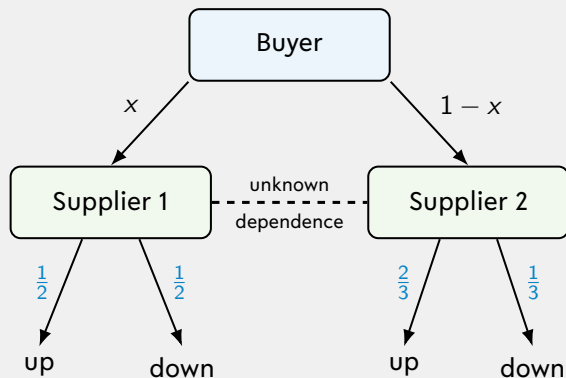
| $p_{\text{ind}}$ | $u_2$         | $d_2$         |
|------------------|---------------|---------------|
| $u_1$            | $\frac{1}{3}$ | $\frac{1}{6}$ |
| $d_1$            | $\frac{1}{3}$ | $\frac{1}{6}$ |

# Dual-sourcing procurement with disruption



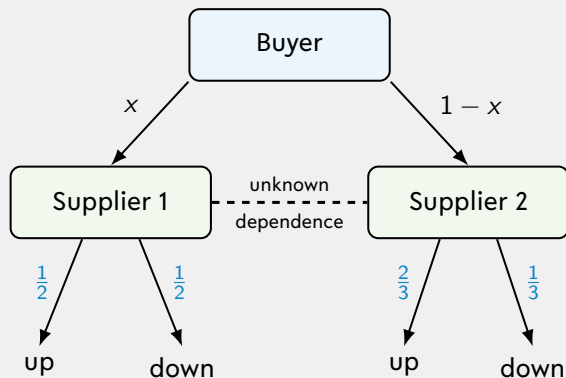
| $p$   | $u_2$             | $d_2$         |
|-------|-------------------|---------------|
| $u_1$ | $\frac{1}{3} + a$ | $\frac{1}{6}$ |
| $d_1$ | $\frac{1}{3}$     | $\frac{1}{6}$ |

# Dual-sourcing procurement with disruption



| $p$   | $u_2$             | $d_2$             |
|-------|-------------------|-------------------|
| $u_1$ | $\frac{1}{3} + a$ | $\frac{1}{6} - a$ |
| $d_1$ | $\frac{1}{3} - a$ | $\frac{1}{6} + a$ |

# Dual-sourcing procurement with disruption



| $p$   | $u_2$             | $d_2$             |
|-------|-------------------|-------------------|
| $u_1$ | $\frac{1}{3} + a$ | $\frac{1}{6} - a$ |
| $d_1$ | $\frac{1}{3} - a$ | $\frac{1}{6} + a$ |

- $a \in [-\frac{1}{6}, \frac{1}{6}]$
- $p_{\text{ind}}$  for  $a = 0$
- $p = p_{\text{ind}} + q$ ,  $q = \begin{pmatrix} a & -a \\ -a & a \end{pmatrix}$

Consider a finite state space  $\Omega = \Omega_1 \times \dots \times \Omega_n$  with prescribed  $p_i \in \Delta(\Omega_i)$ .

## Definition

The *dependence uncertainty set* is

$$\mathcal{P} = (\Omega; p_1, \dots, p_n) := \{p \in \Delta(\Omega) \mid p|_{\Omega_i} = p_i \forall i \in \{1, \dots, n\}\}.$$

$\mathcal{P}$  is a compact convex polytope with  $p_{\text{ind}} := p_1 \otimes \dots \otimes p_n \in \mathcal{P}$ .

► extreme points

► dimension

## Definition

The set of *dependence structures* is

► copulae

$$Q = Q(\Omega) := \{q: \Omega \rightarrow \mathbb{R} \mid \sum_{\omega_{-i}} q(\omega, \omega_{-i}) = 0 \forall \omega_i \in \Omega_i\}.$$



## Definition

The set of *dependence structures* is

► copulae

$$Q = Q(\Omega) := \{q: \Omega \rightarrow \mathbb{R} \mid \sum_{\omega_{-i}} q(\omega, \omega_{-i}) = 0 \forall \omega_i \in \Omega_i\}.$$

## Proposition

Every  $p \in \mathcal{P}$  can uniquely be written as  $p = p_{\text{ind}} + q$  with  $q \in Q$ . Furthermore,

$$\mathcal{P} = \{p_{\text{ind}} + q \mid q \in Q\} \cap \Delta(\Omega).$$

Consider the set  $\mathcal{F}$  of acts  $f: \Omega \rightarrow X$  (Anscombe and Aumann, 1963).  
Today,  $X = \mathbb{R}$  and risk-neutrality, i.e.,  $u(x) = x$ .

# Preferences and Dependence Uncertainty

Consider the set  $\mathcal{F}$  of acts  $f: \Omega \rightarrow X$  (Anscombe and Aumann, 1963).  
Today,  $X = \mathbb{R}$  and risk-neutrality, i.e.,  $u(x) = x$ .

We propose to say a preference relation  $\succeq$  on  $\mathcal{F}$  reflects *dependence uncertainty*, if there are marginals  $p_i \in \Delta(\Omega_i)$  and an act-dependent *ambiguity index*  $\alpha: \mathcal{F} \rightarrow [0, 1]$  such that  $\succeq$  is represented by the functional

$$V(f) = \alpha(f) \min_{p \in \mathcal{P}} \int_{\Omega} f \, dp + (1 - \alpha(f)) \max_{p \in \mathcal{P}} \int_{\Omega} f \, dp,$$

where  $\mathcal{P} = \mathcal{P}(\Omega; p_1, \dots, p_n)$ .

## Definition

An act  $f \in \mathcal{F}$  is called *dependence neutral*, if for all  $p, p' \in \Delta(\Omega)$  with  $p|_{\Omega_i} = p'|_{\Omega_i}$  for all  $i \in \{1, \dots, n\}$  it holds that  $\int_{\Omega} f \, dp = \int_{\Omega} f \, dp'$ . Their set is  $\mathcal{F}^{\text{dn}}$ .

Equivalently,  $\int_{\Omega} f \, dq = 0$  for all  $q \in Q$ .

| $f_1$ | $u_2$ | $d_2$ |
|-------|-------|-------|
| $u_1$ | x     | x     |
| $d_1$ | 0     | 0     |

$$f_1 \in \mathcal{F}_1$$

| $f_2$ | $u_2$   | $d_2$ |
|-------|---------|-------|
| $u_1$ | $1 - x$ | 0     |
| $d_1$ | $1 - x$ | 0     |

$$f_2 \in \mathcal{F}_2$$

| $f_1 + f_2$ | $u_2$   | $d_2$ |
|-------------|---------|-------|
| $u_1$       | 1       | x     |
| $d_1$       | $1 - x$ | 0     |

$$f_1 + f_2 \in \mathcal{F}^{\text{dn}}$$

- A1.  $\succsim$  is complete, transitive and non-trivial.
- A2. ( $\mathcal{F}^{\text{dn}}$ -Archimedean) For all  $g \in \mathcal{F}$  and all  $f, h \in \mathcal{F}^{\text{dn}}$ , if  $f \succ g \succ h$ , then there exist  $\lambda, \kappa \in (0, 1)$  such that  $\lambda f + (1 - \lambda)h \succ g \succ \kappa f + (1 - \kappa)h$ .

- A3. ( $\mathcal{F}^{\text{dn}}$ -Independence) For all  $f, g, h \in \mathcal{F}^{\text{dn}}$ ,  $\lambda \in (0, 1)$

$$f \succsim g \iff \lambda f + (1 - \lambda)h \succsim \lambda g + (1 - \lambda)h.$$

- A4. ( $\mathcal{F}^{\text{dn}}$ -Dominance) For all  $f \in \mathcal{F}$ ,  $g \in \mathcal{F}^{\text{dn}}$

$$f(\omega) \succeq (\preceq) g(\omega) \forall \omega \implies f \succsim (\preceq) g.$$

## Theorem

A preference relation  $\succsim$  on  $\mathcal{F}$  satisfies *Axioms 1–4* if and only if there exist an ambiguity index  $\alpha: \mathcal{F} \rightarrow [0, 1]$  and  $p_1, \dots, p_n \in \Delta(\Omega_1), \dots, \Delta(\Omega_n)$  such that  $\succsim$  is represented by the functional

$$V(f) = \alpha(f) \min_{p \in \mathcal{P}} \int_{\Omega} f \, dp + (1 - \alpha(f)) \max_{p \in \mathcal{P}} \int_{\Omega} f \, dp,$$

where  $\mathcal{P} := \mathcal{P}(\Omega; p_1, \dots, p_n)$ .

# Compatibility to other uncertainty representation

- Maxmin expected utility (MEU) (Gilboa and Schmeidler, 1989, JME)

$$V_{\text{MEU}}(f) = \min_{p \in \mathcal{C}} \int_{\Omega} f \, dp \quad \text{where } \mathcal{C} \subseteq \mathcal{P}.$$

# Compatibility to other uncertainty representation

- Maxmin expected utility (MEU) (Gilboa and Schmeidler, 1989, JME)

$$V_{\text{MEU}}(f) = \min_{p \in \mathcal{C}} \int_{\Omega} f \, dp \quad \text{where } \mathcal{C} \subseteq \mathcal{P}.$$

- Variational preferences (V) (Maccheroni, Marinacci, and Rustichini, 2006, ECMA; Hansen and Sargent, 2001, AER)

$$V_{\text{VP}}(f) = \min_{p \in \Delta\Omega} \left\{ \int_{\Omega} f \, dp - c(p) \right\} \quad \text{where } c(p) = \infty \text{ on } \Delta(\Omega) \setminus \mathcal{P}.$$



# Compatibility to other uncertainty representation

- Maxmin expected utility (MEU) (Gilboa and Schmeidler, 1989, JME)

$$V_{\text{MEU}}(f) = \min_{p \in \mathcal{C}} \int_{\Omega} f \, dp \quad \text{where } \mathcal{C} \subseteq \mathcal{P}.$$

- Variational preferences (V) (Maccheroni, Marinacci, and Rustichini, 2006, ECMA; Hansen and Sargent, 2001, AER)

$$V_{\text{VP}}(f) = \min_{p \in \Delta\Omega} \left\{ \int_{\Omega} f \, dp - c(p) \right\} \quad \text{where } c(p) = \infty \text{ on } \Delta(\Omega) \setminus \mathcal{P}.$$

- Smooth ambiguity (Klibanoff, Marinacci, and Mukerji, 2005, ECMA)

$$V_{\text{smooth}}(f) = \int_{\Delta(\Omega)} \phi \left( \int_{\Omega} f \, dp \right) \, d\mu \quad \text{where } \text{supp}(\mu) \subseteq \mathcal{P}.$$

## Dependence sensitive act

| $f$   | $u_2$ | $d_2$ |
|-------|-------|-------|
| $u_1$ | 1     |       |
| $d_1$ |       |       |

## Dependence sensitive act

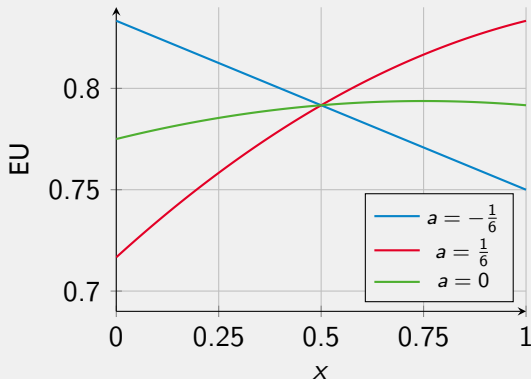
| $f$   | $u_2$                    | $d_2$                    |
|-------|--------------------------|--------------------------|
| $u_1$ | 1                        | $x + \frac{1}{2}(1 - x)$ |
| $d_1$ | $\frac{1}{2}x + (1 - x)$ |                          |

## Dependence sensitive act

| $f$   | $u_2$                    | $d_2$                                          |
|-------|--------------------------|------------------------------------------------|
| $u_1$ | 1                        | $x + \frac{1}{2}(1 - x)$                       |
| $d_1$ | $\frac{1}{2}x + (1 - x)$ | $\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$ |

# Dependence sensitive act

| $f$   | $u_2$                    | $d_2$                                          |
|-------|--------------------------|------------------------------------------------|
| $u_1$ | 1                        | $x + \frac{1}{2}(1 - x)$                       |
| $d_1$ | $\frac{1}{2}x + (1 - x)$ | $\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$ |

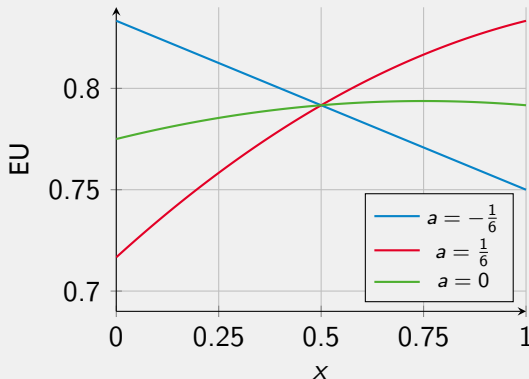


► state-wise utility

# Dependence sensitive act

| $f$   | $u_2$                    | $d_2$                                          |
|-------|--------------------------|------------------------------------------------|
| $u_1$ | 1                        | $x + \frac{1}{2}(1 - x)$                       |
| $d_1$ | $\frac{1}{2}x + (1 - x)$ | $\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$ |

- $a = -\frac{1}{6} \rightarrow x^* = 0$
- $a \in \{0, \frac{1}{6}\} \rightarrow x^* = 1$
- $a \in [-\frac{1}{6}, \frac{1}{6}] \& \text{ MEU} \rightarrow x^* = \frac{1}{2}$



► state-wise utility

## Humans neglect dependence

- when making financial decisions (Eyster and Weizsäcker, 2010; Eyster and Weizsäcker, 2016; Kallir and Sonsino, 2009),
- when forming beliefs (Enke and Zimmermann, 2019, REStud).

## Humans neglect dependence

- when making financial decisions (Eyster and Weizsäcker, 2010; Eyster and Weizsäcker, 2016; Kallir and Sonsino, 2009),
- when forming beliefs (Enke and Zimmermann, 2019, REStud).

## Formal representation

$$V(f) = \int_{\Omega} f \, dp_{\text{ind}}.$$

- We make dependence neglect testable by an axiom!



A5. (Subspace Separation) For all  $f_i, g_i \in \mathcal{F}_i$ ,  $y \in X$ , non-null  $E_{-i} \subseteq \Omega_{-i}$ ,

$$f_i \succsim g_i \iff f_{i[E_{-i}]}y \succsim g_{i[E_{-i}]}y.$$

|                                                                                                                                                                                                                                                                                                                     |               |       |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-------|-------|---|-----|-------|---|-----|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-------|-------|-----|-----|-------|-----|-----|
| <table border="1" style="border-collapse: collapse; margin: 0 auto;"> <tr><td><math>f_1</math></td><td><math>u_2</math></td><td><math>d_2</math></td></tr> <tr><td><math>u_1</math></td><td>1</td><td>1</td></tr> <tr><td><math>d_1</math></td><td>0</td><td>0</td></tr> </table>                                   | $f_1$         | $u_2$ | $d_2$ | $u_1$ | 1 | 1   | $d_1$ | 0 | 0   | $\succsim$ | <table border="1" style="border-collapse: collapse; margin: 0 auto;"> <tr><td><math>g_1</math></td><td><math>u_2</math></td><td><math>d_2</math></td></tr> <tr><td><math>u_1</math></td><td>0.8</td><td>0.8</td></tr> <tr><td><math>d_1</math></td><td>0.4</td><td>0.4</td></tr> </table>                               | $g_1$         | $u_2$ | $d_2$ | $u_1$ | 0.8 | 0.8 | $d_1$ | 0.4 | 0.4 |
| $f_1$                                                                                                                                                                                                                                                                                                               | $u_2$         | $d_2$ |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $u_1$                                                                                                                                                                                                                                                                                                               | 1             | 1     |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $d_1$                                                                                                                                                                                                                                                                                                               | 0             | 0     |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $g_1$                                                                                                                                                                                                                                                                                                               | $u_2$         | $d_2$ |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $u_1$                                                                                                                                                                                                                                                                                                               | 0.8           | 0.8   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $d_1$                                                                                                                                                                                                                                                                                                               | 0.4           | 0.4   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $\iff$                                                                                                                                                                                                                                                                                                              |               |       |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| <table border="1" style="border-collapse: collapse; margin: 0 auto;"> <tr><td><math>f_{1[u_2]}y</math></td><td><math>u_2</math></td><td><math>d_2</math></td></tr> <tr><td><math>u_1</math></td><td>1</td><td><math>y</math></td></tr> <tr><td><math>d_1</math></td><td>0</td><td><math>y</math></td></tr> </table> | $f_{1[u_2]}y$ | $u_2$ | $d_2$ | $u_1$ | 1 | $y$ | $d_1$ | 0 | $y$ | $\succsim$ | <table border="1" style="border-collapse: collapse; margin: 0 auto;"> <tr><td><math>g_{1[u_2]}y</math></td><td><math>u_2</math></td><td><math>d_2</math></td></tr> <tr><td><math>u_1</math></td><td>0.8</td><td><math>y</math></td></tr> <tr><td><math>d_1</math></td><td>0.4</td><td><math>y</math></td></tr> </table> | $g_{1[u_2]}y$ | $u_2$ | $d_2$ | $u_1$ | 0.8 | $y$ | $d_1$ | 0.4 | $y$ |
| $f_{1[u_2]}y$                                                                                                                                                                                                                                                                                                       | $u_2$         | $d_2$ |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $u_1$                                                                                                                                                                                                                                                                                                               | 1             | $y$   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $d_1$                                                                                                                                                                                                                                                                                                               | 0             | $y$   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $g_{1[u_2]}y$                                                                                                                                                                                                                                                                                                       | $u_2$         | $d_2$ |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $u_1$                                                                                                                                                                                                                                                                                                               | 0.8           | $y$   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |
| $d_1$                                                                                                                                                                                                                                                                                                               | 0.4           | $y$   |       |       |   |     |       |   |     |            |                                                                                                                                                                                                                                                                                                                         |               |       |       |       |     |     |       |     |     |

A5. (Subspace Separation) For all  $f_i, g_i \in \mathcal{F}_i$ ,  $y \in X$ , non-null  $E_{-i} \subseteq \Omega_{-i}$ ,

$$f_i \succsim g_i \iff f_i[E_{-i}]y \succsim g_i[E_{-i}]y.$$

### Theorem

*If  $\succsim$  is MEU with prior set  $\mathcal{C} \subseteq \mathcal{P}(\Omega; p_1, \dots, p_n)$ , Axiom 5 holds if and only if  $\succsim$  is SEU with respect to the independent product  $p_{\text{ind}} = p_1 \otimes \dots \otimes p_n$ .*

## Definition

We say that  $\succsim$  is *more dependence averse* than  $\succsim'$  if for all  $f \in \mathcal{F}$  and  $g \in \mathcal{F}^{\text{dn}}$

$$f \succsim g \implies f \succsim' g.$$

## Definition

We say that  $\succsim$  is *more dependence averse* than  $\succsim'$  if for all  $f \in \mathcal{F}$  and  $g \in \mathcal{F}^{\text{dn}}$

$$f \succsim g \implies f \succsim' g.$$

## Definition

The *dependence premium* of  $f \in \mathcal{F}$  is the value  $\text{DP}(f)$  satisfying

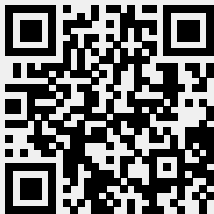
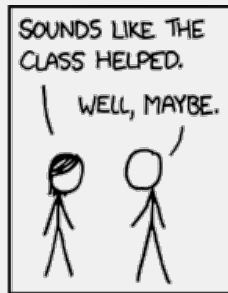
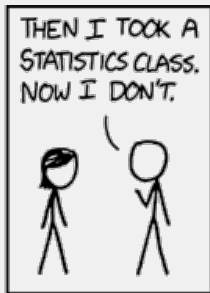
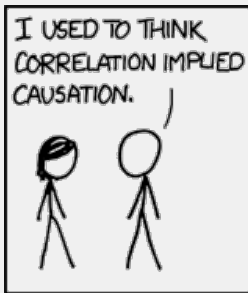
$$\int_{\Omega} f - \text{DP}(f) \, dp_{\text{ind}} = V(f)$$

## Proposition

For dependence uncertainty preferences,  $\succsim$  is more dependence averse than  $\succsim'$  iff one of the following statements hold:

- (i)  $p_i = p'_i$  for all  $i \in \{1, \dots, n\}$  and  $\alpha(f) \geq \alpha'(f)$  for all  $f \in \mathcal{F} \setminus \mathcal{F}^{\text{dn}}$ .
- (ii)  $\text{DP}(f) \geq \text{DP}'(f)$  for all  $f \in \mathcal{F}$ .
- (iii)  $\mathcal{C}' \subseteq \mathcal{C}$  (MEU)
- (iv)  $c^* \leq c^{*'} (VP)$
- (v)  $\phi = h \circ \phi'$ ,  $h$  concave (smooth)

► plots



Bauch and Hartmann, 2026

# Thank You!

-  Anscombe, Francis J and Robert J Aumann (1963). “A definition of subjective probability”. In: *The Annals of Mathematical Statistics* 34.1, pp. 199–205.
-  Bauch, Gerrit and Lorenz Hartmann (2026). *Dependence uncertainty: a decision-theoretic approach*. arXiv: 2503.13416 [econ.TH].
-  Enke, Benjamin and Florian Zimmermann (2019). “Correlation neglect in belief formation”. In: *The review of economic studies* 86.1, pp. 313–332.
-  Eyster, Erik and Georg Weizsäcker (2010). *Correlation neglect in financial decision-making*. DIW Berlin Discussion Paper.
-  — (2016). “Correlation neglect in portfolio choice: Lab evidence”. In: *Available at SSRN 2914526*.
-  Gilboa, Itzhak and David Schmeidler (1989). “Maxmin expected utility with non-unique prior”. In: *Journal of Mathematical Economics* 18.2, pp. 141–153.

-  Hansen, Lars Peter and Thomas J Sargent (2001). “Robust control and model uncertainty”. In: *American Economic Review* 91.2, pp. 60–66.
-  Kallir, Ido and Doron Sonsino (2009). “The neglect of correlation in allocation decisions”. In: *Southern Economic Journal* 75.4, pp. 1045–1066.
-  Klibanoff, Peter, Massimo Marinacci, and Sujoy Mukerji (2005). “A smooth model of decision making under ambiguity”. In: *Econometrica* 73.6, pp. 1849–1892.
-  Maccheroni, Fabio, Massimo Marinacci, and Aldo Rustichini (2006). “Ambiguity aversion, robustness, and the variational representation of preferences”. In: *Econometrica* 74.6, pp. 1447–1498. ISSN: 00129682, 14680262. URL: <http://www.jstor.org/stable/4123081>.
-  Nelsen, Roger B (2006). *An introduction to copulas*. Springer. ISBN: 978-0-387-28659-4. DOI: 10.1007/0-387-28678-0.



## Theorem

*The following are equivalent for a probability  $p \in \mathcal{P}$  and any divergence  $D(\cdot \parallel \cdot)$  with infinitely decreasing slope in 0.*

- (i)  $p$  is an extreme point of  $\mathcal{P}$ .*
- (ii)  $p$  is maximally zero.*
- (iii)  $p$  is a strict local maximizer of the divergence  $D(\cdot \parallel p_{\text{ind}})$ .*

E.g.,  $\varphi$ -divergence:  $\varphi'(0) = -\infty$ . Examples:  $\alpha$ -divergence for  $\alpha \in [0, 1]$ , including (generalized, reversed) Kullback-Leibler divergence, Jeffrey's divergence, Jensen-Shannon divergence, squared Hellinger distance, Neyman  $\chi^2$ -divergence (aka reverse Pearson).

Rectangular operations are dependence structures  $q$  with support on a rectangle:

$$\begin{array}{ccc}
 (\omega_i^*, \omega_{-i}^*) & \xleftarrow{a} & (\omega_i^*, \omega_{-i}^\dagger) \\
 & \text{X} & \\
 (\omega_i^\dagger, \omega_{-i}^*) & \xrightarrow{a} & (\omega_i^\dagger, \omega_{-i}^\dagger)
 \end{array}$$

## Proposition

Rectangular operations form a spanning system of  $Q$ . Setting  $m_i := \# \text{supp}(p_i)$ , the dimension of  $\mathcal{P}$  is equal to  $\prod_{i=1}^n m_i - \sum_{i=1}^n (m_i - 1) - 1$ .

- Correlation is often measured by a *copula* in statistics, cf. Nelsen, 2006.
- A copula is a joint distribution of  $[0, 1]^n$  with *uniform* marginals.
- Copula theory works only for real-valued random variables
  - Dependence structures work on arbitrary state spaces.
  - Copulae can indeed be identified as special dependence structures.

▸ dependence structures

# Dependence Structures and Copulae — II

| $C$           | $\frac{2}{3}$     | 1             |
|---------------|-------------------|---------------|
| $\frac{1}{4}$ | $\frac{1}{6} + x$ | $\frac{1}{4}$ |
| 1             | $\frac{2}{3}$     | 1             |

| $C - C_{\text{ind}}$ | $\frac{2}{3}$ | 1 |
|----------------------|---------------|---|
| $\frac{1}{4}$        | $x$           | 0 |
| 1                    | 0             | 0 |

| $p_{\text{ind}} + q$ | $\frac{2}{3}$     | 1                  |
|----------------------|-------------------|--------------------|
| $\frac{1}{4}$        | $\frac{1}{6} + x$ | $\frac{1}{12} - x$ |
| 1                    | $\frac{1}{2} - x$ | $\frac{1}{4} + x$  |

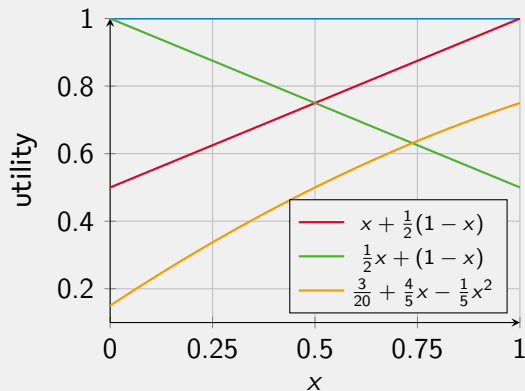
| $q$           | $\frac{2}{3}$ | 1    |
|---------------|---------------|------|
| $\frac{1}{4}$ | $x$           | $-x$ |
| 1             | $-x$          | $x$  |

A copula  $C$  and its corresponding dependence structure  $q$  for  $\Omega_1 = \{\frac{1}{4}, 1\}$ ,  $\Omega_2 = \{\frac{2}{3}, 1\}$  with  $x \in [-\frac{1}{6}, \frac{1}{12}]$ .

► dependence structures

## Dependence sensitive act – State-wise Utility

| $f$   | $u_2$                    | $d_2$                                          |
|-------|--------------------------|------------------------------------------------|
| $u_1$ | 1                        | $x + \frac{1}{2}(1 - x)$                       |
| $d_1$ | $\frac{1}{2}x + (1 - x)$ | $\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$ |



► back

# Dependence Premia for Different Preferences

